Chapter 7

Designing survey experiments☆

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Abstract
A survey experiment is an experiment whose randomly assigned treatment is administered within the context of a survey. Survey experiments are attractive because they give researchers tight control over the details of implementation. The key downside of survey experiments is the artificial setting, which makes it difficult to establish clear links between the theoretical model, experimental design, and real-world phenomenon of interest. In this chapter, we discuss the design of survey experiments through the lens of their fundamental advantages and disadvantages. We first discuss the problem of selecting treatments that operate only via intended mechanisms. We then discuss the selection of outcome measures, emphasizing the desirability of specificity, real-world costs or stakes, and testing for persistence. We then consider issues that affect statistical power, including covariate adjustment, multi-item scales, and attention and quality checks. After briefly discussing sample representativeness, we conclude by summarizing our discussion through a set of recommendations.

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1 Introduction

Surveys are a popular tool for data collection across the social sciences. A survey experiment is an experiment whose randomly assigned treatment is administered within the context of a survey. Background variables and outcome measures are also usually closely tied to the survey, but can also be collected from other sources, e.g. administrative data that can be linked to survey respondents. The fact that treatment is administered within the survey means that survey experiments can generally be fielded more quickly and at lower cost than field experiments, in which treatment is administered in a more naturalistic setting.

In addition to their relative ease of implementation, survey experiments are attractive because they give researchers tight control over the details of implementation, particularly the administration of treatment. In observational research, treatment exposure is non-random, so estimating unbiased treatment effects requires either very strong assumptions or a complete accounting of all other potential sources of outcome differences across treatment groups. Field experiments randomly assign treatment but generally leave some room for selection in and out of treatment, also known as non-compliance. In contrast, survey experiments give the researcher full control over how subjects are actually treated: Provided that the survey is implemented properly, all subjects who are assigned to a given treatment will in fact be exposed to it. Similarly, survey experiments give researchers precise control over how outcome measures are constructed and measured.

The main downsides of survey experiments are the difficulty of (1) clearly linking the treatment to the theoretical model and (2) selecting outcome measures that capture the real-world phenomenon of interest.[1] For linking treatments to theory, the essential task is to assure that treatments operate only through the hypothesized mechanism and not through other mechanisms. This requires the researcher to specify a causal model, consider alternative causal pathways, design and select treatments that target the only desired causal pathway, and include outcome measures that verify that (un)desired pathways were (not) manipulated. When selecting outcomes, the artificial nature of surveys can be partially overcome by selecting measures that are precise, costly, have real-world stakes, or mirror real-world choices. Examining whether effects are persistent can also be important for understanding how treatments work.

Following our discussion of designing treatments and outcome measures, we turn to considerations for the analysis of survey experiments, all of which require planning during the design phase. We discuss statistical power and design features that can improve it, such as controlling for pre-treatment variables, attention checks, and using multiple measures of outcomes. We also consider the issue of representative samples, including the conditions under which representativeness is substantively important and the practical tradeoffs associated

[1] Concerns like these are often discussed through the lens of internal and external validity (McDermott, 2011).

with using weighted estimators. We conclude with a list of recommendations for practice.

Our chapter complements other recent reviews of experimental design, such as Haaland and colleagues' (2023) article on designing information provision experiments. Relative to Haaland and colleagues, we provide a more general framework for the selection of treatments and outcomes, more detailed advice on design and analysis practices that enhance statistical power, and a more general discussion of representativeness that is agnostic as to the population of interest. In contrast, Haaland and colleagues provide more detail on the measurement of beliefs and specific threats to inference that arise in the information provision context, and discuss representativeness through the lens of making inferences about the adult population in the United States.

2 Treatments, stimuli, and mechanisms

Perhaps the key issue for scholars using survey experimental approaches is to clearly link the operationalized survey experiment to their theoretical model. Social scientists are already inclined to specify causal models prior to thinking about empirical tests. However, there can be slippage when moving from a theoretical model to an empirical one, making empirical tests less dispositive than they seem at first glance.

To make this discussion more concrete, consider a researcher who is studying attitudes toward legal immigration. The researcher suspects that if the public had more accurate beliefs about immigrants' skill levels, support for immigration would change. To test this hypothesis using an externally valid stimulus, the researcher randomly assigns some subjects to read a short news article containing information about educational attainment and job skills among legal immigrants, as well as some contextual information like background on the immigration system and some of the arguments for and against it. Following the stimulus, the researcher asks subjects a question designed to measure support for immigration. (We discuss the selection of outcome measures in the next section.)

In this example, what's the treatment? The stimulus is a news article, which may affect support for immigration through one of several mechanisms: beliefs about immigrant skill levels, beliefs about the immigration system in general, and arguments for and against various immigration policies (Fig. 1). If the researcher defines the treatment as a news article about legal immigration, the existence of more than one possible mechanism is not a problem: each component is just part of the total treatment effect of the article. However, if the researcher defines the treatment as beliefs about immigrant skill levels, the existence of the other mechanisms is a problem: the effects of the stimulus are not completely attributable to the factor that the researcher defines as treatment.

The more general problem is that whatever causal factor the researcher wants to manipulate must be targeted by the experimental stimulus. If the stimulus exactly matches the researcher's definition of the treatment—in this case, the effect

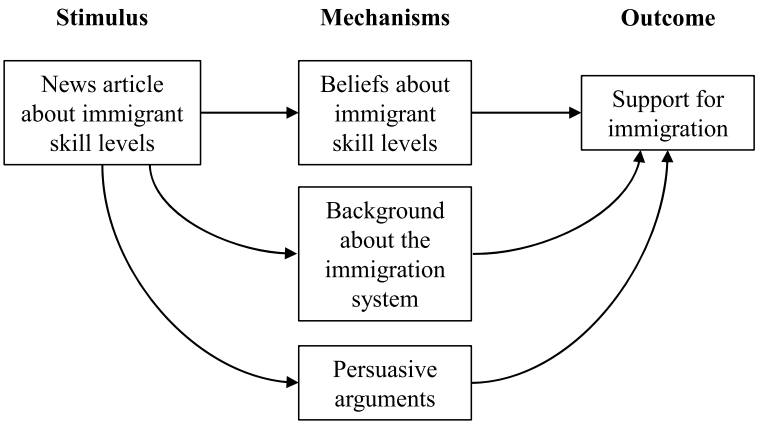

FIGURE 1 Potential mechanisms for the effect of a news article about immigration on support for immigration. Depending on the researcher's definition of the treatment, the various mechanisms may either be viewed as "part of the bundle" or as threats to inference.

of a newspaper article—a diversity of possible mechanisms is not a problem, so long as all the possible mechanisms are tightly linked to the theoretical construct that the stimulus intends to approximate. However, researchers are often interested in defining treatment in a way that is less directly tied to the nature of the stimulus. The looser the match between the stimulus and the intended treatment mechanism, the greater the burden on the researcher to demonstrate that the stimulus affects outcomes **only** through the mechanism that is being defined as the treatment. This is analogous to the burden that users of instrumental variables face when defending the exclusion restriction assumption: the exogenous shock (in this case, the randomly assigned stimulus) must only affect outcomes through the treatment.

The challenge of selecting a stimulus that only affects outcomes through the purported pathways arises in many contexts. For example, consider a treatment that manipulates (beliefs about) a candidate for office's race. This treatment may affect other beliefs about the candidate as well, such as their likely partisanship, issue positions, or issue prioritization. Similarly, a treatment designed to manipulate a neighbor's partisanship may change assessments of their religion, issue commitments and values, and even preferred sports and hobbies. A treatment designed to manipulate the nationality of a product's manufacturer may alter beliefs about the product's quality, cost, or carbon footprint.

In this way, even apparently simple treatments may be "bundles" that affect outcomes through multiple pathways. For example, researchers sometimes use treatments that are newspaper articles (e.g., Fowler and Margolis, 2014) or multiple sets of facts about a topic (e.g., economic inequality). In these cases, the researchers have to make assumptions about the mechanisms by which these treatments operate and assume that the treatments do not perturb other dimensions/mechanisms that could also affect the outcome. If a researcher's interest is

in information about economic inequality, incidental features of a news article like background about the tax code may be problematic. But if one's interest is in news exposure or the effect of a particular genre of news article (such as op-eds (Coppock 2023) or fact-checks (Porter and Wood 2021)), the same features are arguably part of the theoretical construct. Whether a particular dimension is an appropriate feature of the construct is ultimately a substantive question about what constitutes a theoretically well-defined treatment.

2.1 Assuring that treatments work as intended

To verify that treatments work through the intended mechanisms, we recommend that researchers include outcome measures that should be affected if those mechanisms are operative. Continuing the earlier example, if the treatment is defined as beliefs about immigrant skill levels, it is desirable to show that the treatment generates differences in beliefs about immigrant skill levels.[2] Comparing average beliefs across treatment conditions provides concrete evidence that the stimulus affects the construct that is defined as the treatment. Additionally, if an experiment produces unexpected findings, including such outcome measures helps rule out the possibility that individuals simply interpreted the treatments differently than anticipated.[3] In contrast, a researcher who does not measure beliefs about immigrant skill levels and fails to find any effect on support for legal immigration has no way to determine whether the null result is attributable to a failure to manipulate beliefs or to a lack of a relationship between beliefs and attitudes (a refutation of a theory).

Theories also frequently make predictions regarding moderators, which are characteristics of respondents or treatments that are expected to predict larger or smaller effects (i.e., the characteristic "moderates" the magnitude of the treatment effect). When theories make such predictions, a common approach for estimating heterogenous treatment effects is to collect pre-treatment measures of the moderating characteristic, then check whether effects are truly larger or smaller where predicted. For example, Hainmueller and Hiscox (2010) use a simple model of labor market competition to predict that less-educated workers will be relatively more opposed to low-skilled immigration, while highly educated workers will be more opposed to high-skilled immigration. To test this, they randomly assign respondents to answer questions about high- and low-skilled immigrants. They then interact treatment assignment with indicators for the respondent's education level (low or high) and jointly test the statistical significance of the interaction terms. They do not find the expected heterogeneity

[2] We focus here on average evaluations, but interventions may also affect beliefs about variance, and variance may be consequential for understanding choice. See Leemann et al. (2021) for a discussion of strategies for measuring belief distributions in surveys.

[3] Moreover, a researcher may be wrong and pre-treatment beliefs may actually be more optimistic than supposed, in which case a treatment designed to move beliefs in one direction may be moving them in the opposite direction.

in effects, suggesting that non-economic factors may be needed to explain voter attitudes.

Importantly, randomization of treatment does not ensure randomization of any moderator/covariate. That is, any pre-treatment characteristics may be correlated with another pre-treatment characteristic that could also predict treatment heterogeneity, so the key advantage of experimentation (all else equal by design) is lost. For example, consider again Hainmueller and Hiscox's test of the labor market competition model. Compared to low skill workers, high skill workers may, on average, be more conservative or Republican, have fewer immigrant family members, form their beliefs about appropriate immigration levels on the basis of different values, or have more accurate ex ante beliefs about immigrant skill levels. Each of those factors might explain a smaller (or bigger) reaction to the treatment. This means that simply estimating a treatment × worker skill interaction may produce a biased estimate of the causal effect of worker skill on treatment effectiveness because worker skill is correlated with other factors that also moderate that treatment effect. This is the same broad problem of omitted variables bias in any observational study and the fact that analysis centers on an experiment does not obviate this concern.[4]

As such, it is useful to approach causal claims regarding mediation or moderation with skepticism. When examining heterogeneous effects according to moderators, we suggest carefully examining the robustness and sensitivity in much the same way that one probes the robustness of observational claims. For example, is the estimated effect of a treatment times covariate term robust to including other relevant treatment by covariate interactions? More generally, it is important for the researcher to be transparent that this component of their analysis is observational in nature, despite being embedded in a larger experimental design.

Just as researchers may want to find evidence that particular mechanisms are operating, they may also want to rule out one or more causal pathways. On the front end, the crucial step is to design treatment arms that attempt to hold the undesired pathway constant (Dafoe et al., 2018).[5] In the news article example, if it is not feasible to remove persuasive arguments or background about the legal immigration system, researchers could take a "leave one out approach," including persuasive arguments and background information in all treatment conditions and randomly assigning only whether the stimulus includes factual information about immigrant skill levels. Similarly, in the candidate race example, the researcher can specify characteristics of the politician that they do not intend to manipulate. For another example, Kim (2023) studies the effect of "rags-to-riches" television programming on beliefs about economic mobility. Such programming could affect beliefs about mobility through its effect on beliefs about how the economy works, or through some non-economic pathway

[4] By contrast, if a researcher can randomly manipulate a moderator, then doing so provides causally informative evidence about its effect.

[5] This is also one advantage of multifactorial experiments in which multiple factors are manipulated simultaneously (further discussion appears below in Selecting Valid Outcome Measures).

like general optimism. To rule out general optimism, Kim's control condition consisted of optimistic programming that was not about economic mobility.

Despite one's best efforts, one may still worry (or worry that others will worry) that a treatment affects outcomes through alternative pathways. To verify that a design has successfully ruled out a given causal pathway, we suggest including outcome measures that will allow testing for those effects. To understand whether the treatment avoids affecting mechanisms that the researcher intends to exclude, outcomes corresponding to those mechanisms can be measured in each treatment arm (see Gerber et al. (2017) for such tests in the specific context of beliefs about immigration). If there are no differences, while not dispositive (in that there could be offsetting effects), it helps rule out the possibility of alternative causal pathways. Differences across treatment arms, in contrast, would suggest that the exclusion restriction is not met. Returning to the earlier example (Fig. 1), perhaps the researcher took pains to make sure that the treatment does not include persuasive arguments about immigration or background about the mechanics of the legal immigration system; the success of these efforts could be tested with outcome measures related to these same factors. This strategy can also be applied to concerns about researcher demand effects, that is, that treatment effects are an artifact of respondents recognizing the purpose of the study and trying to give helpful responses (Mummolo and Peterson, 2019). By including outcome measures that test respondents' beliefs about the purpose of the study, researchers can learn whether a precondition for such effects (recognition of the purpose of the study) is present.

Another example of this approach appears in Orr and Huber (2020). Earlier research had randomly assigned partisan labels to hypothetical strangers and interpreted the resulting increases in animosity as evidence for group identity-based theories of partisanship, as opposed to policy-based theories. Orr and Huber (2020) show that when the stranger's policy positions are also provided, effects of partisanship on animosity are much less pronounced.[6] Further, Orr et al. (2023) find that partisan labels have large effects on beliefs about the policy positions someone holds. These findings suggest that the causal pathway linking outgroup partisanship to animosity is largely not partisanship *per se*, but instead what partisanship communicates about existing value and policy commitments. Here, we once again return to the key question of what constitutes treatment. If one views partisanship as a bundle of identity- and policy-based mechanisms, the distinction between the effect of partisan affiliation itself and the effect of partisan affiliation on beliefs about issue positions is irrelevant. However, if one is interested in distinguishing group-based partisan animosity from substantive conflict about policy, a treatment that perturbs both is less informative.

It is also possible that in addition to the treatment stimulus, other features of the survey affect how respondents understand the treatment. For example, suppose one is interested in whether information discussing the history of the First

[6] By contrast, Orr and Huber (2020) show providing information about partisanship does not substantially diminish the effect of policy disagreement on animosity.

Amendment affects support for allowing peaceful protests on college campuses. If subjects are asked prior to treatment about antisemitism or the KKK, they may think of those sorts of protestors when answering the outcome question rather than whatever types of protestors would naturally have come to mind. The simplest way to deal with such concerns is to increase the distance between the two measures, e.g. by measuring the covariates in a separate pre-treatment survey. If a pilot study is being conducted, one could also randomly assign the presence or absence of the potentially problematic questions and include outcome measures designed to test for priming effects, e.g. an open-ended question about the types of protestors the respondent was thinking about.

All told, our advice for assuring that treatments work as intended can be boiled down to a few recommendations:

- At the design stage, attempt to craft treatment conditions that only manipulate the intended factors. As part of this process, try to anticipate unintended causal pathways and assure that they are held constant across treatment conditions.
- Include outcome measures that can be used to assess whether various mechanisms—intended or unintended—were operative. This includes theoretical predictions as well as differential attrition (discussed below).
- Include pre-treatment measures that can be used to verify theoretical predictions about heterogeneous effects, i.e. moderators. Be aware that between-group differences in treatment effects are not causally identified and require the same care as any observational study.

2.2 Differential attrition or breakoff

In addition to worrying about the various theoretical factors that a treatment could affect, researchers must also consider how treatments affect respondents' willingness to complete the survey.[7] If a treatment makes respondents more or less likely to complete the survey—on average or differentially across subgroups—it may no longer be credible to assume that the experiment provides an all-else-equal comparison of outcomes across treatment arms. This problem is known as differential breakoff (when it takes place during the survey) or differential attrition (when it takes place between waves of a panel survey). From a design standpoint, researchers can minimize the risk by assuring that each treatment arm takes about the same time to complete and requires about the same cognitive burden. In the news article example, rather than skip straight to the outcome measures, control respondents could be assigned to read a text of similar length and difficulty about an unrelated topic.[8] On the analysis side, researchers can test for the problem by regressing an indicator for attrition or breakoff on

[7] The problem of attrition has historically been labeled "survey mortality."

[8] To provide assurance that idiosyncratic features of the placebo do not drive the estimates, researchers can also select several placebos and randomize between them (Porter and Velez, 2022). If a pure control condition is also included, one can compare the placebos to this baseline.

treatment indicator(s). To test for differences across subgroups, researchers can regress an attrition or breakoff indicator on all treatment-by-covariate interactions, then test the individual or joint significance of the interactions using an F-test.

Importantly, it is possible to accidentally sweep differential attrition under the rug during data collection. Survey vendors may obscure breakoff as part of their quality control, e.g. by only providing data on complete responses or by assigning additional respondents to treatment conditions with greater attrition (Bailard et al., 2023). Researchers who manage their own data collection may also accidentally drop cases of breakoff, e.g. by downloading data from Qualtrics before incomplete responses are closed or failing to ensure that enumerators return partially complete responses.

3 Selecting valid outcome measures

While survey experiments have the advantage of tight control and rapid data collection, there is nonetheless a critical challenge that is common to all analysis of survey data: What does a survey response actually mean? In this section, we link this critique to research on the nature of survey responses, which holds that surveys generally measure on-the-spot thinking rather than fixed beliefs. We then consider how researchers can improve the match between these artificial choices and real-world decisions in terms of specificity, stakes, and persistence. We also consider the pernicious possibility that a treatment alters the way individuals interpret a survey item without changing underlying beliefs or preferences.

As a starting point for these issues, it is useful to have in mind a model of survey responses. Among political scientists and survey researchers, the most influential accounts hold that survey responses are constructed on the fly (Zaller, 1992; Tourangeau et al., 2000). That is, (most) individuals do not carry around in their heads preformed answers to questions like "What do you expect the rate of [inflation/deflation] to be over the next 12 months?" (Manski, 2018) or "in your view, should immigration be kept at its present level, increased or decreased?" (Gallup 2024). Instead, when confronted with such a question, individuals take a "sample" of the information, experiences, values, etc. stored in their memory and make an on-the-spot judgment about which response option best represents their beliefs. At best, one hopes that the on-the-spot choice that a respondent makes in a survey is the same as the choice they would make if faced with the same situation in the real world, e.g. when considering whether to buy a product based on its price or when evaluating a political candidate's proposal to increase or decrease legal immigration.

At the heart of the three issues we consider in this section—specificity, stakes, and persistence—is the fundamental issue of how the inferential situation created by a survey relates to choices in the real world. Contrast the ambiguity of a survey response with a realized choice—that is, a revealed preference. For example, suppose a person chooses to buy product A given its price rather than

product B or nothing at all. By definition, it follows that the utility of A (given its cost) exceeds both the utility of B (given its cost) and the utility of not spending the money on either. But whereas a physical item is concrete and tangible, survey items are often vague, meaning that they can be interpreted in different ways across individuals or over time. Additionally, survey responses are not behaviors, meaning they are (largely) costless. Finally, just as behavioral choices do not tell us why someone chooses A over B, survey items do not provide direct evidence of why someone provides response like "inflation is too high." The advice that follows focuses on writing questions that effectively measure the construct the researcher is interested in, anticipating objections from skeptical readers, and thinking about measurement error concerns.

3.1 Specificity

Start with the issue of a vague question. For example, the standard General Social Survey (GSS) question about support for redistribution:

> *Some people think that the government in Washington ought to reduce the income differences between the rich and the poor, perhaps by raising the taxes of wealthy families or by giving income assistance to the poor. Others think that the government should not concern itself with reducing this income difference between the rich and the poor. Here is a card with a scale from 1 to 7. Think of a score of 1 as meaning that the government ought to reduce the income differences between rich and poor, and a score of 7 meaning that the government should not concern itself with reducing income differences. What score between 1 and 7 comes closest to the way you feel?*

Note the multiple ways in which this item is vague. Begin with the response options. If 1 means "the government ought to reduce the income differences between rich and poor" and 7 means "the government should not concern itself with reducing income differences," what's a 3, and how is that different from a 2 or a 4? Such a concern matters both when we compare responses across individuals and over time, because people may interpret the items differently or the same person may interpret it differently from one day to the next. To the degree this is simply random measurement error it may be of little concern (other than inducing attenuation), but what if it isn't random error?

Turning to the stem of the question, note the multiple and vague issues that it raises: 1) Who is rich and who is poor? 2) How will income differences be reduced, via taxes or spending or both, 3) If taxes, what taxes, how much, and on whom? 4) If spending, what programs, how much, and to whom? Once again, this vagueness might not be much of a problem if it is a simple issue of random error but consider it in the context of a survey experiment where a treatment may influence preferences toward redistribution. Suppose a treatment presents information about income for those at the 5th and 95th percentiles of the income distribution, thereby making poverty and inequality apparent. This treatment

might operate by changing beliefs about the earnings of the poor or by changing beliefs about inequality (and, per our earlier argument, we recommend measuring both). More perniciously, it might have no effect on those beliefs but might instead simply change who counts as rich and who counts as poor. Note also that such concerns are not resolved by asking the same person the same question twice (a within-person design) because there is no assurance that the item would be interpreted in the same way over time.

The solution to vague items is specificity. For example, imagine rewriting the GSS question as follows:

- "Some people are concerned about income inequality in the United States and believe the Federal Government ought to do more to reduce it, while others are less concerned. Which of the following steps would you support the federal government taking to reduce income inequality?"
- "Increasing federal income taxes on everyone making more than [$100,000 (the top X% of all earners), $250,000 (the top X% of all earners), $500,000 (the top X% of all earners), $1 million (the top X% of all earners)] by about [1, 2, 3, 4, 5]% of their income each year"
- "Providing a monthly payment of [$100,$500,$1000] cash to all families with children whose household earning are less than [$15,000 (the bottom X% of all families), $25,000 (the bottom X% of all families), $50,000 (the bottom X% of all families)]"

On the one hand, this level of specificity requires greater attention to detail and may lead to concerns that respondents do not actually carry around preferences or opinions about the items being asked about. Our view based on theories of the survey response is that these concerns apply equally to less specific items. Given this, specific items are more interpretable across- and within-individual (and more likely to hold constant meaning across treatment conditions), providing a more concrete sense of what people are supposed to be making an inference about. Nonetheless, these items are not perfect. For example, if one proposes raising taxes on different groups by different amounts, one may also perturb beliefs about the groups to which such revenue will be redistributed and by how much.

The specificity and real-world relevance of choices can be further enhanced through multi-factorial vignette designs linked to relevant outcomes. Multi-factorial vignettes randomize several attributes of a stimulus at once to allow estimating the influence of the attributes on an outcome of interest rather than asking respondents to state their preferences directly.[9] For example, to understand support for offering citizenship to individual immigrants, Hainmueller and Hopkins (2014) designed a choice experiment that closely resembles the

[9] An additional advantage of this approach, noted above, is that by specifying and manipulating those additional factors, one can "hold them constant," addressing concerns that manipulation of one dimension affects inferences about another dimension that affects choice. As we note above, one can also test for such spillover inferences by randomizing not just the level, but also the presence, of additional information.

real-world choice in Swiss votes on admitting new migrants (see Hainmueller and Hangartner 2013). In the experiment, they randomize multiple attributes of pairs of fictional immigrants and examine the effects of those characteristics on whether a person is admitted. They use the term "conjoint" to describe this paired design where individuals choose between candidates in a pair. Importantly, the effects they estimate for each manipulated dimension (e.g., country of origin) on support for admission in this design are after holding constant all other factors they manipulate (e.g., gender, education, profession, language skill, etc.), but may also reflect inferences on dimensions that are affected by one manipulation but which are not specified (e.g., country of origin may affect beliefs about religion).

In general, we see five key advantages to multi-factorial vignette designs. First, the measured outcome can often be stylized to resemble a real-world outcome. For example, Orr and Huber (2020) ask respondents to create a seating chart for dinner party guests with randomly assigned characteristics. Second, the fact that the researchers manipulate multiple features can obscure the researcher's theoretical interest, reducing concerns about social desirability bias and demand effects. Third, because many features are manipulated, one can compare the magnitude of effects across the different randomized features. For example, how does the effect of information about the prevalence of immigrant groups vary based on country of origin (Newman and Malhotra, 2019)? Fourth, one can assess how effects of one trait vary with the provision of other information. For example, how does the effect of undemocratic positions covary with policy differences between candidates (Graham and Svolik, 2020)? Finally, these designs can often be repeated multiple times within the same survey, improving statistical power by increasing the number of observations per respondent (Bansak et al., 2018).

3.2 Stakes

Specifying a clear inferential target is progress but does not fully address the question of whether survey responses would predict real behavior. It is at least possible with specific questions to identify a relevant behavioral benchmark: would a person vote in support of a federal referendum measure adopting the specific policy they proposed supporting, for example. But how does one deal with the cheap talk problem? How can one turn survey responses into revealed preferences? This question is likely to depend on what one believes is the source of measurement error in stated versus revealed preferences.

If measurement error is simply a product of cheap talk, then the key goal is to create incentives that encourage people to respond based on their genuine beliefs or preferences. For example, Bullock et al. (2015) and Prior et al. (2015) show that the degree of partisan differences on survey items asking about politically-relevant facts is reduced when individuals are paid for correct answers. Bullock et al. (2015) extend this design based on a model of the survey response that

shows that when expressing support for one's party is enjoyable, payment increases the probability of a sincere response by boosting the value of honesty. Importantly, that model also shows that paying people to "get it right" will have no effect on partisan difference when individuals are least informed—because when people have little basis to make an inference about which response option is more likely to be correct, payment for correct answers has a minimal effect on the expected value of choosing one option or another. Based on the model, Bullock and colleagues extend the design to include payment for "don't know" response options.[10]

If measurement error comes from a more fundamental disconnect between stated and revealed preferences, one can devise costly or quasi-behavioral measures. For example, if people are making forecasts about the future economy, one can make a future state-dependent payoff to them, so that they will earn more if their forecast is correct. One challenge is that to make those future payments large enough to make them meaningful without exhausting the researcher's resources (and, as we note above, it is also useful to think about the problem of what one would expect if individuals know they don't know). Here are a few other examples of how to collect behavioral responses:

- Ask respondents to choose how to allocate a bonus payment between themselves and a third party, making the revealed preference a meaningful measure (Lee, 2022).
- Give respondents an option to choose between a personal payment or allocating that payment to a third party. Several studies allow respondents to allocate money between themselves and a charitable donation (Bakaki and Bernauer, 2016; de Wit and Bekkers, 2020; Goff et al., 2017; Grigorieff et al., 2020). In the same spirit, Elías et al. (2019) offer respondents a $1 bonus if they choose to let the researchers donate $1 to a charity that supports organ donation, meaning that those who choose not to donate must also forego personal gain.
- Paying respondents to make accurate forecasts about the future, e.g. the number of COVID-19 cases (Allcott et al., 2020) or whether Donald Trump would manage to retain power after losing the 2020 election (Graham and Yair, 2024).
- Dellavigna et al. (2017) use the fact that taking surveys is time consuming to show that some respondents were so motivated to communicate their participation that they choose to take a much longer survey merely to have the chance to tell the researchers they had voted.
- To study factors that affect willingness to read fact-checks, Graham and Porter (2025) allow respondents to choose whether to be redirected to a fact-checking website after the conclusion of the survey.
- Ask survey respondents to sign a public petition (Grigorieff et al., 2020) or send a letter to an elected official (Adida et al., 2018; Barton and Pan, 2022).

[10] For a more extensive discussion of how to use incentives to accurately assess beliefs, see Hill (2017).

3.3 Persistence

Even if survey measures have a concrete real-world referent and real payoffs or stakes, a concern is that the effect on such choices may be fleeting. In survey experiments that measure persistence, effects often decay quickly. At best, effects persist at roughly one third to one half of their original size in surveys conducted one to four weeks after the original study (Dowling et al., 2020; Jungherr et al., 2021; Porter and Wood, 2021; Coppock, 2023; Coppock et al., 2023). In many cases, researchers find little or no evidence of persistence (e.g., Druckman and Nelson 2003; Mutz and Reeves 2005; Bailard et al. 2023).

What are the implications of the weak persistence of survey experimental treatment effects? One possibility is that humans forget. Told by a doctor at an annual checkup to get enough sleep and exercise, we dutifully do so the weekend after our visit. But the next weekend comes, and we've forgotten the doctor's advice and are staying up late and skipping the gym. Similarly, treatment effects—like the effects of providing information about the level of inequality in society—may dissipate because people simply forget that information.

Our earlier discussion of models of survey response points to another, more pernicious possibility: despite the change in survey responses, the underlying beliefs or attitudes may never have changed. Recall that in our summary of theories of the survey response, respondents were described as drawing on considerations that are accessible. If someone reads about poverty before answering a question about economic redistribution, it is possible that the treatment may cause them to rely more on their underlying feelings about poverty, and less on beliefs about economic efficiency, when making their on-the-spot judgment about whether redistribution is desirable—even though the underlying beliefs never changed. This type of effect is known as a priming effect: treatments make making some considerations more readily available without altering underlying beliefs.[11]

There are two ways to think about a lack of persistence, and neither is clearly the only correct answer. One possibility is that this shows true "steady-state" beliefs and opinions are hard to change and so it may be that survey experiments just tend to overstate true treatment effects. This is a pessimistic account. A second perspective is that the stimulus in a survey experiment is just one of many relevant pieces of information that people encounter as they move through their daily lives. People may also hear similar information about the issue from media, advertisements, friends, or at church. All these stimuli may affect salience or attitudes in a small way. This means that a "one-off" survey experiment provides a reasonable snapshot of a reaction to a single "dose" of the treatment, but

[11] Priming is related to the concept of framing. Framing is usually understood as affecting which values are relevant for a choice. For example, framing the choice about preferred levels of redistribution as about degree of care for the needy rather than economic efficiency may increase reported support for redistribution without actually changing preferences about the tradeoff between care and efficiency.

the effects should be expected to decay as memory fades and other information affects beliefs and salience. Outside of the survey context, the equivalent real-world treatment would be steady state exposure to information about income inequality/etc.

Beyond this debate about what persistence means, it also seems clear that it is important to assess it. This might include conducting panel surveys in which subjects are reinterviewed at points in time distant from an initial treatment, as well as designing surveys so that outcomes are measured at varying distances from treatment within the same survey. When a treatment effect dissipates over time, researchers can still be sure that the treatment changes survey responses. Depending on the context, this may or may not rule out the causal mechanism that interests the researcher. Importantly, differential attrition between survey waves causes the same problems in estimates of persistence that we discuss above, and should be handled in the same manner.

4 Statistical power

Another attractive feature of survey experiments is the researcher's control over factors that affect statistical power, i.e. the probability of rejecting the null hypothesis of no effect when an effect exists. Most obviously, with an unlimited budget, a researcher could simply choose an arbitrarily large sample size to detect even modest effect sizes with high probability. Of course, budgets are not unlimited. Given this, researchers have developed other routes to maximizing statistical power. In this section, we discuss covariate adjusted estimators, using multi-item scales as outcome measures, attention and quality checks, and power calculations.

4.1 Covariate adjustment

The most straightforward way to estimate the average treatment effect in an experiment is to calculate a difference in means. This can be accomplished by fitting the regression $Y_i = \beta_0 + \beta_1 D_i + \epsilon_i$, where Y_i is the outcome and D_i is an indicator for the randomly assigned treatment. In most cases, however, the difference-in-means approach unnecessarily leaves statistical power on the table.

A better approach to estimating the average treatment effect is to use a covariate-adjusted estimator,[12] an approach that Zhang and Bungi also discuss in Chapter 10. Specifically, as part of the regression specification used to estimate the average treatment effect, we recommend controlling for pre-treatment variables that are correlated with the outcome. This reduces the standard error

[12] Discussions of field experiments often discuss blocked random assignment, i.e. assigning treatment to fixed proportions of respondents within subgroups defined by covariates. This has similar benefits to covariate adjustment but the need to impose subgroup-based quotas on the random assignment can be difficult to implement on most survey platforms. If blocking is possible in your setting, see Gerber and Green (2012).

of the treatment effect estimate by "soaking up" unexplained variation in the outcome. However, since treatment is randomly assigned, there is no systematic relationship between treatment status and variables that were measured prior to the treatment. Except in very small samples, controlling for pre-treatment variables does not change the point estimate, only the standard error.[13] These properties make covariate adjustment an indispensable tool in the analysis of experiments, in and outside of surveys. For example, the Food and Drug Administration recommends the use of covariate adjustment in clinical trials (FDA, 2023). To maximize the benefits of covariate adjustment, researchers can turn to Lin's (2013) estimator, which involves some additional preprocessing of the data but guarantees that the adjusted estimate will have a smaller standard error than the unadjusted estimate (Ye et al., 2023).

Adjusting for a pre-treatment measure of the outcome

The most effective approach for improving statistical power is to include a measure of the outcome in the pre-treatment portion of the survey, then control for it in the statistical analysis. That is, one can estimate $Y_{i,t} = \beta_0 + \beta_1 D_i + \beta_2 Y_{i,t-1} + \epsilon_i$, where $Y_{i,t-1}$ is the pre-treatment measure of the outcome. In addition to or instead of the pre-treatment measure of the outcome, researchers can also control for other pre-treatment variables that are correlated with the outcome.

Covariate adjustment is related to first-differencing the outcome measure, an approach that is perhaps more intuitive but is inferior from a statistical power perspective. The first-differencing strategy is to subtract the pre-treatment measure of the outcome from the post-treatment measure; that is, to fit $Y_{i,t} - Y_{i,t-1} = \beta_0 + \beta_1 D_i + \epsilon_i$. Simple algebra shows that this is equivalent to the approach in the previous paragraph with the constraint that $\beta_2 = 1$. Letting OLS choose the value of β_2 is more statistically efficient because OLS chooses the value that minimizes the amount of unexplained variation in the outcome. Additionally, to allow more functional form flexibility, particularly with categorical data, $Y_{i,t-1}$ can be replaced with indicators for each level, reducing concerns that the mapping between pre and post responses may be non-linear. Pedagogically, however, first-differencing is an intuitive entry point because it makes clear that incorporating covariates reduces variance in the outcome measure (e.g., see Gerber and Green 2012, Chapter 4.1 and 4.2).[14]

A potential downside of measuring the outcome pre-treatment is consistency pressure, wherein the act of answering the same question earlier in the survey attenuates treatment effects by "anchoring" respondents to their response to earlier questions. In a rare test of this common suspicion, Clifford et al. (2021),

[13] More technically, covariate adjusted estimators are consistent, with a small bias that quickly converges to 0 as the sample size increases (Lin, 2013).

[14] Another benefit of including a pre-treatment measure of the outcome is to allow a more informed debate about whether the treatment alters the composition of a sample either demographically or in a way correlated with the outcome. In the income inequality example, if there is concern that the treatment causes respondents who are opposed ex ante to redistribution to drop out of the survey at higher rates, one can use the pre-treatment outcome measure to test for this selection problem.

who randomize the inclusion of pre-treatment outcome measures in otherwise identical experiments. Our interpretation of their results is that large amounts of attenuation are unlikely. However, based on a post hoc power calculation (Appendix A), our assessment is that their design was under-powered to detect more modest amounts of attenuation (40 percent or less) that may be substantively important in some contexts. Cautious researchers can take steps like placing pre-treatment measures of the outcome as far away as possible from the post-treatment measure, e.g. early in the survey or in a separate survey wave. Pre-treatment measures can also be presented in a slightly different format, which is unlikely to substantially attenuate the correlation between the two measures but may make the purpose of the question less obvious to the respondent.

The above discussion assumes that even though the outcome is measured pre-treatment, the researcher conducts a randomized experiment prior to the main outcome measure, re-measuring the control outcome for some respondents and measuring the treatment outcome for others. To gain additional statistical power, researchers may be tempted to instead "treat everyone" at the second stage. In these situations, the benefits of randomization are lost. The within-subject estimate of the causal effect is only identified by the assumption that in the absence of treatment, $Y_{i,t} = Y_{i,t-1}$. Without a control group, one cannot test this assumption.[15] However, if this assumption does not hold, a treatment that on average perturbs Y over time cannot be distinguished from any other pattern that induces a change in Y. Even though statistical precision can be gained by treating everyone, it comes at the expense of admitting bias.

Adjusting for other prognostic covariates

Researchers who cannot, or choose not to, include a pre-treatment measure of the outcome can still reap the benefits of covariate adjustment if they have access to other pre-treatment variables that are correlated with the outcome. Such measures are often referred to as "prognostic covariates." Because their correlation with the outcome is likely to be weaker than a pre-treatment measure of the outcome itself, the power gains from controlling for other prognostic covariates are not likely to be as great. However, since covariate adjustment is essentially unbiased in large samples (see footnote 13) choosing not to adjust for prognostic covariates amounts to sacrificing statistical power for nothing. We recommend that researchers control for prognostic covariates when they are available, even if a pre-treatment measure of the outcome has also been collected.

In this more general setting, researchers should be mindful of three potential pitfalls. First, because researchers can generally choose from a range of covariates, it is possible to use covariate adjustment in an unprincipled manner, engaging in specification search until one finds a set of covariates that (by

[15] For example, imagine a simple scenario in which people inflate (deflate) their report about a behavior when first asked, but then adjust downward (upwards) their report when asked again. In this case, the actual estimate of change in beliefs will be confounded by this degree of change in reporting.

chance) move the point estimate in a convenient direction. To avoid this, researchers can pre-register their covariate adjustment strategy and exactly follow their pre-registered plan during analysis. For example, they can specify the exact variables they will adjust for and a regression equation they will estimate, or specify an algorithm like fitting a LASSO on the control group (Bloniarz et al., 2016). In addition to improving the integrity of the estimates, preregistration builds credibility, allowing researchers to prove that covariate adjustment was an *ex ante* decision to improve statistical power. When preregistration is absent, reviewers and other readers can reasonably wonder whether covariate adjustment was an *ex post* decision, perhaps designed to deal with randomization failure or as a result of specification search.

Second, as a general rule, researchers should only control for pre-treatment variables, that is, variables that are measured prior to treatment. If post-treatment variables are affected by the treatment, including them as controls can introduce bias and the direction of the bias is ambiguous (Montgomery et al., 2018). Some argue that it is reasonable to control for post-treatment covariates that are arguably unaffected by treatment, e.g. demographic information (Klar et al., 2020).[16] However, other research finds that self-reports of even seemingly static traits like race and ethnicity can be affected by political factors (Egan, 2020). A preregistered adjustment strategy that only makes use of pre-treatment variables allows researchers to rule out bias based only on assumptions that are true by design.

Third, especially when several pre-treatment covariates are used for adjustment, missing data in the covariates may cause researchers to drop respondents who could otherwise be included in the analysis, reducing statistical power and changing the population being studied. This may not be obvious to the researcher when it is happening (e.g., the `lm()` function in R does not explicitly warn researchers that observations with missing data have been dropped). To avoid this problem, researchers should instead impute missing values of the pre-treatment covariates. One option is to impute the missing covariates based on other pre-treatment covariates, and another is to simply impute the mean. Alongside imputation, one can also include indicators for missing values. Since the purpose of the covariates is to reduce the amount of unexplained variation in the outcome, concerns about the accuracy of the chosen imputation method are secondary to ensuring that all respondents are included in the analysis.

4.2 Multi-item scales as outcome measures

Another route to increasing statistical power is to combine multiple measures of the outcome into a single index, e.g. by taking a simple average or using factor analysis. For example, to measure support for redistribution, one could measure support for several redistributive tax proposals rather than just one proposal,

[16] Indeed, one argument for measuring certain covariates post-treatment is that measuring them pre-treatment can affect the treatment itself. See above.

then take the average of these measures. (In Chapter 1, Snowberg and Yariv call this approach "quasi-repetition".) Under a model in which each item reflects the construct of interest plus random noise, scales like this are thought to reduce measurement error (Ansolabehere et al., 2008). In general, measurement error in an outcome variable that is uncorrelated with the independent variable increases the standard error of estimates (Wooldridge, 2012, 318).

Multi-item scales can also complement covariate adjustment. Relative to individual items, multi-item indices are more highly correlated over time (Ansolabehere et al., 2008). This means that when the outcome variable is measure as a multi-item scale, it will be more highly correlated with the pre-treatment measure. Recall that covariate adjustment is more effective when the correlation between the covariates and the outcome is higher. By increasing the correlation between pre- and post-treatment measures of the outcome, multi-item scales help researchers get more out of covariate adjustment. Because the gains are proportional to the square of the inverse of the correlation, gains can still be substantial when the measured characteristic is highly stable over time, e.g. political partisanship (Broockman et al., 2017).

4.3 Attention and quality checks

Survey-takers do not always pay careful attention to the treatment or outcome questions. If one imagines that each response is a mixture of the true value and random noise (in proportion to the amount of attention paid), inattentiveness attenuates treatment effect estimates toward zero (Kane, 2024). The resulting bias reduces statistical power. Researchers can deal with the problem of inattentiveness by including measures of attention. The appropriate use of these measures depends on whether they appear before or after the treatment and whether inattentive subjects are part of a population of interest.

If attention is measured pre-treatment, researchers can drop respondents who fail some number of checks (Berinsky et al., 2014) or examine how the treatment effect varies according to the level of attention (Kane, 2024). Some survey platforms allow researchers to screen out low-quality respondents[17] or to request additional respondents based on quality indicators. The downside of this approach is that it may affect representativeness. Less attentive respondents may be systematically different than more attentive respondents and may also be inattentive when they encounter the real-world equivalent of the treatment. To the extent this is the case, attenuation due to inattention can be thought of not as a bias, but as making the resulting treatment effect estimates more representative of the real-world population or treatment effect. Consistent with the discussion above, this is partly a question of how the researcher defines treatment: is treatment reading a news article or having the opportunity to read a news article?

[17] For ethical reasons, it is common practice that questions that are used to rule respondents ineligible should be included at the beginning of the survey. Otherwise, respondents will not be compensated for their time.

Pre-treatment attention checks come in two basic forms. Instructional checks start out like a regular survey question but then instruct respondents to provide a response that would be unusual. For example, one of the checks in Berinsky et al. (2014) starts out like a standard question about one's level of interest in politics but directs respondents to select both "very interested" and "extremely interested." Since selecting multiple responses is unusual on such questions, respondents who select both options are assumed to be attentive. Mock vignette checks are designed to mimic the content of the experimental stimulus but are followed by factual questions about the content of the mock vignette (Kane et al., 2023). Respondents who can correctly answer factual questions about the mock vignette are presumed to be more likely to pay attention to the experimental vignette. Because attention may vary over the course of the survey, it is possible that generally attentive respondents may miss an attention check in one instance or that generally inattentive respondents may get one right. Berinsky et al. (2021) recommend using four checks.

Post-treatment attention checks, often referred to as manipulation checks, ask respondents about the details of their experimental condition. For example, Press et al. (2013) randomly assign information about a nuclear attack's severity and check their respondents' level of attention to the treatment by asking an outcome question about the attack's severity. The proportion of respondents who answer correctly, or the treatment effect on the check (if it is constructed in a way that is plausibly symmetrical across conditions), can be used to diagnose how closely respondents attended to the treatment's content (Kane and Barabas, 2019). However, because check passage may be correlated with treatment assignment, researchers should not drop respondents who fail post-treatment checks (Montgomery et al., 2018). For example, Press et al. (2013) dropped respondents who failed their post-treatment manipulation check, resulting in biased estimates of the treatment effect (Aronow and Miller, 2019).

Response time is another indicator of attention. The traditional logic is that "speeders" who quickly complete a survey are unlikely to be paying careful attention (Zhang and Conrad, 2014). Variation in response times can also be an indicator: Read et al. (2022) identify a "slow inattentive" subgroup of respondents who take an unusually long time on some survey questions. Using response time to screen out respondents is risky because if the treatment affects how long the survey takes, response time is partly a post-treatment variable (and so excluding observations on this basis may induce a form of researcher-induced differential attrition). This is especially likely if treated respondents see more content than control subjects (e.g. additional text or images). We advise that researchers who use response time as an indicator of attention limit their focus to response time on pre-treatment questions.

Researchers can also borrow quality check strategies that were originally developed for other purposes. The literature on measuring rare traits finds that respondents who inconsistently report their demographic information, or who identify themselves as having implausible traits (e.g. the ability to name every

Senator who has ever lived, eating concrete for its iron content), are likely to provide inaccurate responses to other questions as well (e.g., gargling bleach to cure covid; Litman et al. 2023). Similarly, Graham (2024) finds that about 2 percent of respondents frequently switch tabs or browser windows during the pre-treatment portion of a survey, suggesting a lack of engagement. Although there is not direct evidence that treatment effect estimates are attenuated among respondents who fail such checks, our sense that any screener that successfully identifies low-quality respondents is likely to be useful in the context of survey experiments with the aforementioned caveat that screening out respondents may also affect representativeness.

4.4 Calculating statistical power

We argue above that large samples, covariate adjustment, multi-item scales, and attention or quality checks can all help increase statistical power—that is, they increase the probability of rejecting the null hypothesis assuming that it is actually false. But how does one decide whether a particular set of design choices has adequate statistical power? Researchers can use simulation or analytic formulas to assess statistical power for a range of possible combinations of design choices, then pick a design that is optimal given their resource or implementation constraints. This is especially valuable in survey experiments given the researcher's unusual level of control over the design. If power for a particular set of design choices is low for effect sizes that a researcher finds substantively interesting, the researcher can try to identify ways to increase power before spending precious time and funding. If power is high, researchers can enjoy the opportunity to make precise inferences about the effect's magnitude or make money-saving design choices like cutting sample size or fielding a one-shot survey instead of a panel.

To calculate statistical power, researchers use simulation or analytic methods to approximate the sampling distribution that is likely be produced by a particular design. Both methods have two fundamental ingredients: (1) an assumption about the true effect size and (2) assumptions about the design factors that determine the standard error, e.g. the sample size, distribution of the outcome measure, and estimator. All statements about power are conditional on these two ingredients. For example, a researcher might say, "this design has 80 percent power to detect an income effect of $100." This means that if the true effect size were $100, we would expect 80 percent of estimates to be far enough from the null hypothesis to reject it. If $100 is a substantively interesting effect size in our context and 80 percent sounds good, a researcher might be reassured by this. If $100 is not interesting or 80 percent sounds risky, the researcher can look for ways to do better.

To understand how a set of assumptions comes together into a statement about statistical power, consider the following example. A researcher expects the effect size to be 1 and plans to test the null hypothesis that the true effect

is 0. They have enough money to pay 2,000 respondents to take a 10-minute survey through an online convenience sample vendor. The experiment has two treatment arms that are independently randomly assigned with equal probability ($p = 0.5$). Based on existing research, they know that the standard deviation of the outcome measure is 11.5, and they don't expect the treatment to affect it. This gives the researcher enough information to calculate power using one of two approaches, simulation or analytic formulas.

To calculate statistical power analytically, the researcher could plug the inputs into an off-the-shelf formula that applies to their case (e.g., Gerber and Green 2012, Appendix 3.1) or piece together a calculation by finding the expected standard error of their estimate, the rejection threshold for their statistical test, and the probability that an estimate is at least this far from the rejection threshold. In this case, the researcher plans to conduct a simple difference-in-means test and reject the null hypothesis if $p < 0.05$ in a two-sided test. They can estimate the standard error of the estimate using the formula $\sqrt{\frac{V_0}{N_0} + \frac{V_1}{N_1}}$ (Gerber and Green 2012, 3.3), where V and N are the variance and sample size in the control and treatment groups (0 and 1); in this case, the standard error is $\sqrt{\frac{11.5^2}{1000} + \frac{11.5^2}{1000}} = 0.51$. They will reject the null hypothesis whenever the estimate is 1.96 standard errors away from 0, which is $1.96 \times 0.51 \approx 1$. If their expectation that 1 is the true effect size is correct, this gives them a 50 percent chance to reject the null: about half the time the estimate will be 1 or greater and the null will be rejected, but the other half of estimates will be less than 1 and result in a chance failure to reject the null. This illustrates a useful rule of thumb: if the expected effect size is equal to 1.96 times the standard error, a two-sided test has only a 50 percent chance to reject the null hypothesis of no effect. Contrary to the instincts one might develop based on significance testing, good power requires an expected effect size of at least three or four times the standard error, not just two. For more on analytic methods and rules of thumb, we recommend Rainey (2024).

To simulate statistical power, researchers write computer programs that simulate the experiment thousands of times, then study its properties. In this case, the researcher would write a computer program that generates data based on a true effect size of 1, then conduct the statistical test and record whether the null hypothesis was rejected. Power is equal to the proportion of simulations in which the null is rejected. Simulation-based approaches are generally more flexible because the researcher can easily account for design choices that off-the-shelf tools or formulas may not capture. Simulation-based approaches also tend to take as their inputs the same factors that the researcher is trying to decide how to manipulate (e.g., the sample size and estimator) while leaving the math to the computer. A complication of simulation-based approaches is the need to simulate the structure of the data. Researchers can sidestep this by using an existing dataset that contains variables that are similar to the intended outcome measure, or (if budget constraints allow) by conducting a pilot study.

Software is also available to assist with the simulation process. For example, the DeclareDesign software asks the researcher to specify a range of inputs such as a sample size, effect size, and estimator, then returns an estimate of statistical power (Blair et al., 2019). For a more detailed introduction to simulation-based power calculations, we recommend Blair et al. (2023, Chapter 10, Sections 3 and 4).

Regardless of whether a researcher calculates power using a simulation or analytic methods, the chief benefit of power calculations is to examine how different features affect a design's ability to detect the effects of interest. In some cases, researchers might find that their design has excellent power (say, 99 percent) to detect the expected effects. Based on this information and their risk tolerance, they could decide to reduce the sample size or add an additional treatment arm designed to test some other component of their theory, re-doing the power calculation to assure that this design choice does not reduce power below the levels they are willing to accept. Alternatively, the researchers might find that their design has poor statistical power (say, 50 percent). In this case, the researchers might decide to reduce the number of treatments, use a more efficient estimation strategy (e.g., covariate adjustment), or shorten the survey and free up money for additional respondents. They could also decide not to run the experiment at all. In short, power calculations give researchers a concrete basis to determine whether their design is adequate, what they need to change to make it adequate, and whether the needed changes are feasible given their constraints.

It is important to note that the metric of statistical power—the percent chance of rejecting the null when it is false—is not designed to inform researchers about how precisely they can estimate the magnitude of an estimate. When researchers care about more than just rejecting the null (for example, if they expect a null and want to precisely estimate that null effect), 80 or 90 percent power might not be sufficient. In this case, researchers can turn to other information about the sampling distribution that is generated as a part of the power calculation. For example, if the researcher extracts the point estimates from their simulations rather than just whether the null hypothesis was rejected, they can construct a simulated sampling distribution and standard error, then assess whether these are narrow enough for the application.

5 Representativeness

So far, we have sidestepped the issue of representativeness. If a sample is representative of a population, the sample's characteristics are equal to the population's characteristics in expectation, differing only due to sampling variability. Although discussions of representativeness often revolve around the adult population of the country, concerns about representativeness are relevant regardless of what population one is studying (people, firms, geographies, etc.).

For survey experiments in particular, the key question about representativeness is whether the effect on the sampled population is equal (in expectation) to

the effect on the population of interest. That an experimental sample is not itself representative of the population of interest is a common complaint. But, it is essential to note that for this complaint to matter, the estimated treatment effect in the sample must be different from the treatment effect in the population, which is only true if there is treatment effect heterogeneity. To the extent that researchers have good reasons to think that treatment effects in their sampled population are similar to the target population, a lack of representativeness is not a concern. Existing research on the generalizability of treatment effects across population suggests reasons for cautious optimism. Treatment effects on college students and MTurk workers are similar to general population effects, even though more-representative samples tend to produce more-representative effects (Mullinix et al., 2016; Coppock et al., 2018; Yeager et al., 2019). Similarly, treatment effects on foreign policy elites can be roughly approximated by studies of ordinary people, especially when ordinary people's responses are weighted to match foreign policy elites on traits like educational attainment (Kertzer, 2022). In sum, representativeness helps, but a lack of representativeness does not mean that all hope is lost.

The first-order method of making a sample more representative is to recruit a more representative sample. Recruitment methods can generally be classified as probability-based or convenience. In a convenience sample, selection into the sample is completely at the respondent's discretion. For example, respondents respond to advertisement for a specific survey or sign up for services that offer compensation for survey-taking. In a probability sample, respondents are contacted at random with known probability from some sampling frame, such as all phone numbers or all addresses in the country, or a list of all publicly traded companies. Although probability-based methodologies assure random recruitment, the response rate to such invitations are not 100 percent. Because some who are contacted do not participate, there is still selection into probability samples. This is true regardless of the studied population. For example, just as surveys that use random digit dialing to study the general population are likely to oversample older people, people with landlines, and those interested in discussing the topic of the study, a study of corporate executives could oversample executives from a particular industry or with a particular management style.

The back-end method of achieving greater representativeness is to calculate weighted estimates. Given a set of characteristics for which the true population distribution is known, weighting algorithms find a set of weights that produce as close a match as possible between the sample's distribution and the population distribution. To the extent that these characteristics predict differences between the sample and population in terms of other variables of interest (e.g. the outcome variable, the treatment effect), the weighted estimates are representative of the target population. An important limitation of weighting is the scarcity of available variables. Researchers can only weight based on characteristics of the population that (1) have a known ground truth and (2) can be reliably measured in a survey. This is why surveys that target the general population all tend to

weight to the same handful of demographics like age, gender, racial or ethnic identification, and (increasingly) educational attainment. Even when good variables are available, weighting has costs. It usually increases the standard error of treatment effect estimates (Miratrix et al., 2018) and can introduce bias in small samples (Franco et al., 2017). Weighting can also function as a researcher degree of freedom that can lead to selective reporting (Franco et al., 2017; Kim and Wong, 2023). Regardless of whether they plan to use weights, researchers should strive to pre-register their intentions.

All of this discussion boils down to four practical takeaways. First, representativeness issues are applicable regardless of the population of interest. Relative to researchers who are studying the general population, researchers studying specialized populations could have an easier or harder time assessing the representativeness of their survey. It all depends on what traits are observable for both the population of interest and in a survey. Second, a lack of representativeness is not necessarily fatal. If there are good reasons to think that treatment effects in the population one can reach are similar effects on to the population of interest, one can still learn from unrepresentative studies. Third, decisions about whether and how to deal with representativeness should consider costs and benefits. On the front end, the monetary costs of intensive recruiting efforts may not be worth it if treatment effects are likely to generalize. On the back end, the statistical power cost of weighting may not be worth it if theoretically relevant factors are not observable in both the survey and the target population. Fourth and finally, researchers should strive to make decisions about weighting ex ante. Otherwise, weighting becomes a researcher degree of freedom that risks producing false positives.

6 Conclusion

Survey experiments allow researchers to exercise tight control over treatment assignment in a low-cost environment. The price of this control is legitimate concerns about the links between the theoretical model, the experimental design, and the real-world phenomenon of interest. In this concluding section, we organize our discussion into a checklist for researchers who are considering implementing a survey experiment.

- *Pay close attention to the connection between the stimulus and the definition of treatment.* The ease of randomly assigning participants to a treatment condition does not spare the researcher any difficulty in defining the treatment. In fact, given the artificial nature of surveys, defining the treatment in a survey experiment may be more difficult than in a field or natural experiment. Researchers should take care to design treatments that only manipulate factors that are consistent with their intended definition of treatment and should define treatment in a way that is consistent with what was manipulated.

- *Use supplemental outcome measures to test assumptions about how the treatment works and does not work.* Audiences will have legitimate questions about how the treatment worked and whether it is consistent with the researcher's account. Researchers should endeavor to anticipate these questions and include supplemental outcome measures that can test whether the treatment worked as intended. This ultimately works in the researcher's favor. When things go as expected, the researcher has a better case that their theory is correct. When things do not go as expected, the researcher has more information about why.
- *Be mindful of attrition and breakoff.* Survey experimental treatments often affect not just the stimulus that the respondent sees, but the amount of time or cognitive effort required to engage with the survey. This can cause more people in one condition or the other to drop out, potentially creating spurious associations between the treatment and outcome. Researchers should design experiments that try to balance time and effort across conditions and should test for differential breakoff between treatment conditions and types of respondents.
- *Strive to select outcome measures that match the specificity and stakes of real-world choices or behaviors.* Although survey experiments are generally at least somewhat artificial, researchers can incorporate design elements that tighten the connection between the outcomes measured in surveys and the real-world phenomenon of interest. Two especially important types of outcomes are (1) those that reveal preferences rather than ask respondents to state them directly and (2) those that have some external cost or benefit, either to the respondent (e.g. money) or to real-world outcomes of interest (e.g. a petition that will actually be sent to Congress).
- *If fleeting effects are a problem, test for persistence.* Researchers may sometimes be content to understand how people reason or make tradeoffs in the moment. In this case, the potential that survey experimental treatment effects are fleeting or a product of priming may not be a problem. However, to the extent that a researcher wants to make claims that a treatment affected respondents' underlying beliefs or altered their behavior in a lasting way, it is important to test for persistence by collecting follow-up measures of the outcome some weeks or months after the treatment is administered.
- *Pre-register an analytic strategy that involves controlling for prognostic covariates.* While a larger sample results in greater statistical power, the benefits of increasing the sample size decrease as the sample size grows larger. Using pre-treatment covariates to soak up unexplained variation in the outcome provides a second route for shrinking standard errors. However, covariate adjustment can also be used in an unprincipled manner. Researchers should preregister the exact procedure for covariate adjustment, either by specifying the particular variables or an automated variable selection procedure (e.g., Bloniarz et al., 2016).

- *Have an explicit strategy for dealing with inattentive respondents.* Inattentive respondents cause attenuation bias but are also real people whose inattention in a survey may be representative of their behavior in other contexts. Researchers should consider the problem of inattention ahead of time. If representativeness is a secondary concern, we recommend dropping respondents who fail pre-treatment attention checks. Post-treatment checks should be included for diagnostic purposes but should not be used to drop respondents.
- *Check statistical power.* Checking statistical power adds upfront costs but ultimately helps researchers make better investments. At minimum, researchers should try to get a sense of what their standard error might be and use rules of thumb to check that power is likely to be at least 80 or 90 percent. We find it useful to remember that with a two-sided test, an effect size of 1.96 standard errors is only 50 percent powered. An expected effect size of three standard errors or more is likely to satisfy many researchers but may not be sufficient in all situations, e.g. when it is important to precisely estimate an effect's magnitude.
- *Be conscious of representativeness.* The two methods for achieving greater representativeness, recruiting a probability sample and using weights, are costly in terms of money and statistical power. However, they are also the surest way to produce representative estimates. Researchers should carefully weigh these tradeoffs prior to fielding the survey and clearly disclose any weighting procedures in a preregistration document.

Although following these recommendations cannot overcome all the limitations of survey experiments, we think that survey experiments that are consistent with these principles constitute a valuable part of the social scientific toolkit.

Appendix A Power in Clifford et al. (2021)

To aid our assessment of existing evidence regarding consistency bias in survey experiments, we conducted a retrospective power analysis of Clifford et al. (2021), who compare effect sizes in designs with and without a pre-treatment measure of the outcome. To conduct the power analysis, we assumed that the true effect of each treatment was equal to the effect in the post-only design (0.104) and that the standard error of the difference in effects would be equal to the standard error in the meta-analysis (0.014). We specified a range of possible amounts of attenuation between 0.0 to 0.1, which are equal to roughly 0 to 100 percent attenuation relative to the (assumed) true effect.

We then used both simulation and analytic methods to calculate statistical power, i.e. the expected percentage of the time that the estimate of the amount of attenuation would be statistically significant. In each simulation, we draw a random estimate from a normal distribution centered at the original point estimate, then conduct a difference in means test standard error. We repeat the simulations for many possible amounts of attenuation. For the analytic calculation, we use the normal CDF to calculate the proportion of estimates that

would be expected to exceed the rejection thresholds, again for many possible amounts of attenuation. Our code is available in the Harvard Dataverse at https://doi.org/10.7910/DVN/ELDNTA.

The results appear in Fig. 2. In order to reach 80 percent power, the true effect size would need to be about 0.04, about 39 percent of the true effect. At low but plausibly important rates of attenuation, between 10 and 25 percent of the true effect size, power is less than 50 percent. In our assessment, this indicates that the design was well-powered to detect large amounts of attenuation but cannot rule out modest amounts of attenuation that could still be considered substantively significant.

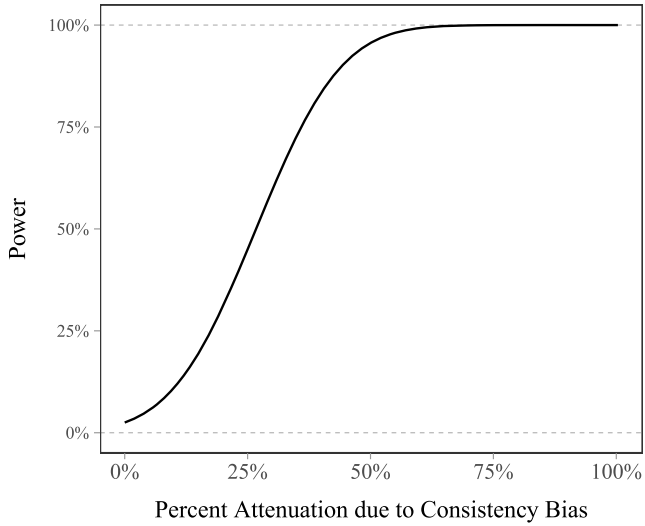

FIGURE 2 Retrospective analysis of statistical power in Clifford et al. (2021). The x-axis is the assumed percent attenuation relative to the originally reported standard effect size without a pre-treatment measure of the outcome. The y-axis is the percentage chance of rejecting the null hypothesis of no attenuation in the design with a pre-treatment measure, based on the originally reported standard error.

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